

WinInteropUtils

WinInteropUtils is a C# library written in .NET 8 that provides managed P/Invoke wrappers for tons of functions you can use, from reading file attributes to interacting with COM interfaces.

The base namespace is `FireBlade.WinInteropUtils`.

⚠ WARNING

WinInteropUtils is still in beta, so bugs may occur. If you experience any bugs, please report them on the [Issues](#) page.

Features

- [WindowsFile](#) file info wrapper
- Full enums (1600+ values) for [HRESULTs](#) and [Win32 error codes](#)
- Dialog boxes
- Useful utility [User32](#) functions ([SendMessage](#), [SetWindowLongPtr](#)...)
- [Constants](#) and [Macros](#)
- [COM interfaces](#)
- Icon management ([DestroyIcon](#), ...)

Namespace FireBlade.WinInteropUtils

Classes

[EnumExtensions](#)

Provides extension methods for [Enum](#) .

[IconExtensions](#)

Adds extension methods to the [Icon](#)  class.

[Macros](#)

Defines Win32 macros.

[MessageLoop](#)

Provides methods for managing message loops.

[Shell32](#)

Declares P/Invoke functions from [Shell32.dll](#).

[StockIconHelper](#)

Allows you to get a shell stock icon. Unlike the built-in .NET [GetStockIcon\(StockIconId, int\)](#)  or [GetStockIcon\(StockIconId, StockIconOptions\)](#)  functions, this wrapper allows you to enter any icon ID you want.

[TitleBarInfo](#)

Stores data about a window's title bar.

[User32](#)

Declares P/Invoke functions from [User32.dll](#).

[VisualStyle](#)

Provides methods for managing and drawing with visual styles.

[WinConstants](#)

Defines common Win32 constants.

[Window](#)

Provides methods for managing Win32 windows.

[WindowClass](#)

Represents a window class.

[WindowLongPtr](#)

Defines default index values for [SetWindowLongPtr\(int, nint\)](#) and [GetWindowLongPtr\(int\)](#).

[WindowStyles](#)

Represents the various standard styles that can be applied to a window (after the window has been created, these styles cannot be modified, except as documented).

[WindowsFile](#)

Stores data about a Windows file.

[ZOrderWindows](#)

Represents windows used for Z position ordering.

Structs

[CreateStruct](#)

The structure passed to the `lParam` parameter of the `WM_CREATE` message of a window.

[DrawTextAdvancedOptions](#)

Represents a structure that contains extended formatting options

[HResult](#)

Represents a COM error code (`HResult`).

[Shell32.SHFILEINFO](#)

The internal Windows API struct for [SHGetFileInfoW\(string, uint, ref SHFILEINFO, uint, uint\)](#)[↗]. Use with [GetFileInfoEx\(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO\)](#).

[WinMessage](#)

Provides information about a Windows message.

Interfaces

[IHandle](#)

Represents an interface for classes wrapping unmanaged handles.

Enums

[ClassStyle](#)

Represents class styles. Class styles define additional elements of the window class.

[ExWindowStyles](#)

Specifies extended window styles.

[Facility](#)

Represents the facility code of a [HResult](#).

[MessageProcessingLimit](#)

Specifies what message types should be processed by [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#).

[MessageRemoval](#)

Specifies message removal options for [PeekMessage\(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit\)](#).

[Shell32.SHGetFileInfoFlags](#)

Provides flags for the Windows API function [SHGetFileInfoW\(string, uint, ref SHFILEINFO, uint, uint\)](#) [↗].

To use this function, call the [GetFileInfoEx\(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO\)](#) wrapper instead.

[StockIcon](#)

Icon identifiers for the [GetIcon\(StockIcon, StockIconOptions\)](#) function.

[SystemColorId](#)

Defines the identifiers for system colors.

[SystemFontId](#)

Defines the identifiers for system fonts.

[ThemeTextOptions](#)

Represents the options for [DrawThemeText\(Graphics, int, int, string, int, ThemeTextOptions, Rectangle\)](#).

[TitleBarElement](#)

Defines elements on a title bar.

[TitleBarElementState](#)

Defines the states of a title bar element.

[Win32ErrorCode](#)

Defines Win32 error codes.

[WindowSetPosOptions](#)

Window sizing and positioning flags for [SetPos\(Point, Size, WindowSetPosOptions, Window?\)](#).

[WindowShowCommand](#)

Represents the show command for [Show\(WindowShowCommand\)](#) and [ShowAsync\(WindowShowCommand\)](#).

[WindowState](#)

Defines states a window can be in.

[WindowsFileAttributes](#)

Specifies attributes for Windows files.

Delegates

[WindowClass.WndProc](#)

A callback function, which you define in your application, that processes messages sent to a window. The `WndProc` name is a placeholder for the name of the function that you define in your application.